

Foundation (Jalapeno)

- Q1.** Evaluate the expression $2 \cdot 3 + 12 \div 4$ using PEMDAS. (B)
 (A) $8 + 3$ 10
 (B) $6 + 6$
 (C) 9 (C)
 (D) 10 13
 (E) 12 (D)
- Q2.** Translate the phrase '5 more than x' into an algebraic expression. 16
 (A) $x - 5$ (E)
 (B) $x + 5$ 12
 (C) $5x$
 (D) $5 - x$
 (E) $x^2 + 5$
- Q3.** Simplify the expression $3x + 2x + 4$ by combining like terms.
 (A) $5x + 4$
 (B) $5x - 4$
 (C) $3x^2 + 4$
 (D) $x + 4$
 (E) $2x + 4$
- Q4.** Substitute $x = 4$ into the expression $x^2 - 3$ and evaluate the result.
 (A) 13
 (B) $16 - 3$
 (C) 10
 (D) 7
 (E) $4^2 - 3$
- Q5.** The price of a stock rose by $5 \cdot 2 + 12 \div 4$ dollars. What was the total increase?
 (A) 7
- Q6.** A company has $12 \cdot 3 + 20 \div 4$ employees. How many employees does the company have?
 (A) $36 + 5$
 (B) 41
 (C) 38
 (D) 40
 (E) 39
- Q7.** Evaluate the expression $(8 - 4) \cdot 3$ using PEMDAS.
 (A) $4 \cdot 3$
 (B) $12 - 4$
 (C) 12
 (D) 16
 (E) 20
- Q8.** Translate the phrase '7 less than y' into an algebraic expression.
 (A) $y + 7$
 (B) $y - 7$
 (C) $7 - y$
 (D) $y^2 - 7$
 (E) $7y$

Open Ended Questions (FRQ)

- Q9.** The total cost of purchasing a stock is given by the expression $5x + 20$, where x is the number of shares purchased. If an investor purchases 10 shares of the stock, what is the total cost?
- Q10.** The expression $3(2x - 1) + 2(x + 4)$ represents the profit of a business after x years of operation. Simplify the expression and evaluate the profit after 2 years.

Chef's Tips

- Tip 1: Emphasize the importance of following the order of operations (PEMDAS) when evaluating expressions.
- Tip 2: Use real-world examples to illustrate the relevance of variables and expressions in business and finance.
- Tip 3: Encourage students to check their work by plugging in values for the variables and evaluating the expressions.